

Year 7 · Chapter 2 · Angles: answers (teacher copy)

2.01 Naming angles (booklet p. 3)

WE DO examples (teacher)

- Ex 1a: $\angle ABC$ or $\angle CBA$
- Ex 1b: vertex K in the middle
- Ex 2a: $\angle QOR$
- Ex 2b: $\angle QOR$, $\angle ROP$, $\angle QOP$
- Ex 3a: $\angle LON$
- Ex 3b: There are several angles at B, so we need 3 letters

Easy (drill)

- | | | |
|------|----------------------------------|----------------------------------|
| 1. X | 8. W | 15. $\angle QEF$ or $\angle FEQ$ |
| 2. D | 9. D | 16. $\angle SXT$ or $\angle TXS$ |
| 3. D | 10. V | 17. $\angle PXV$ or $\angle VXP$ |
| 4. B | 11. S | 18. $\angle CRP$ or $\angle PRC$ |
| 5. Y | 12. A | 19. $\angle EGZ$ or $\angle ZGE$ |
| 6. N | 13. $\angle VSB$ or $\angle BSV$ | 20. $\angle LPZ$ or $\angle ZPL$ |
| 7. E | 14. $\angle WEF$ or $\angle FEW$ | 21. $\angle DPE$ or $\angle EPD$ |

Medium (drill)

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|--------------|---------------|---------------|
| 1. ZM and ZP | 8. AS and AK | 15. RQ and RY |
| 2. CM and CS | 9. QC and QN | 16. WX and WS |
| 3. NU and NX | 10. PV and PD | 17. TS and TK |
| 4. QY and QH | 11. UL and UY | 18. SZ and ST |
| 5. PV and PZ | 12. YE and YG | 19. WU and WL |
| 6. LE and LM | 13. WL and WE | 20. FY and FQ |
| 7. NE and NU | 14. VZ and VM | |

Challenging

- $\angle LMN$
- $\angle QRS$

Extension

- 10
- $\angle AOB$, $\angle AOC$, $\angle AOD$, $\angle BOC$, $\angle BOD$, $\angle COD$

Exit ticket

- Level 1: $\angle TUV$
- Level 2: U, because it is the point where the arms meet
- Level 3: two angles sharing arm BC

2.02 Measuring and drawing angles

(booklet p. 9)

WE DO examples (teacher)

- Ex 1a: 64°
- Ex 1b: accurate 84° angle
- Ex 2a: 128°
- Ex 2b: accurate 117° angle
- Ex 3a: draw 150° on the other side
- Ex 3b: draw 60° on the other side

Easy (drill)

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|----------------|-----------------|-----------------|
| 1. 155° | 8. 170° | 15. 140° |
| 2. 105° | 9. 15° | 16. 135° |
| 3. 115° | 10. 155° | 17. 85° |
| 4. 95° | 11. 160° | 18. 140° |
| 5. 70° | 12. 170° | 19. 65° |
| 6. 35° | 13. 130° | 20. 40° |
| 7. 145° | 14. 140° | |

Medium (drill)

- | | | |
|-----------|-----------|-----------|
| 1. reflex | 4. reflex | 7. reflex |
| 2. obtuse | 5. right | 8. reflex |
| 3. obtuse | 6. obtuse | 9. reflex |

- | | | |
|------------|------------|------------|
| 10. acute | 13. obtuse | 16. reflex |
| 11. obtuse | 14. reflex | |
| 12. reflex | 15. acute | |

Challenging

- $360 - 195 = 165^\circ$ on the other side
- three-quarter turn

Extension

- 150° and 90°
- 75°

Exit ticket

- Level 1: 58°
- Level 2: accurate 135°
- Level 3: draw 110° the other way

2.03 Classifying angles (booklet p. 15)

WE DO examples (teacher)

- Ex 1a: acute, right, obtuse, straight, reflex, revolution
- Ex 1b: $\angle WIH$, obtuse
- Ex 2a: obtuse
- Ex 2b: 270° , reflex
- Ex 3a: $\angle DRA$ acute (70°), $\angle ARB$ obtuse (110°)
- Ex 3b: No: $90^\circ + \text{more than } 90^\circ$ is already more than 180°

Easy (drill)

- | | | |
|-----------|--------------|------------|
| 1. reflex | 8. acute | 15. obtuse |
| 2. obtuse | 9. reflex | 16. obtuse |
| 3. reflex | 10. reflex | 17. obtuse |
| 4. reflex | 11. straight | 18. obtuse |
| 5. reflex | 12. obtuse | 19. acute |
| 6. right | 13. acute | 20. right |
| 7. reflex | 14. obtuse | 21. acute |

Medium (drill)

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|-------------------------|--------------------------|-----------|
| 1. 225° , reflex | 9. 160° , obtuse | 17. true |
| 2. 305° , reflex | 10. 220° , reflex | 18. false |
| 3. 100° , obtuse | 11. 160° , obtuse | 19. true |
| 4. 175° , obtuse | 12. 145° , obtuse | 20. true |
| 5. 260° , reflex | 13. 205° , reflex | 21. true |
| 6. 215° , reflex | 14. 225° , reflex | 22. true |
| 7. 105° , obtuse | 15. 285° , reflex | 23. true |
| 8. 295° , reflex | 16. 340° , reflex | 24. false |

Challenging

- 120° , obtuse
- Yes, e.g. $60^\circ + 50^\circ = 110^\circ$
- 100° , 125° , 150° , 175°

Extension

- 900°
- 3 revolutions

Exit ticket

- Level 1: acute
- Level 2: reflex; straight
- Level 3: $\angle DRA$ acute; $\angle ARB$ obtuse

2.04 Complementary and supplementary angles (booklet p. 21)

WE DO examples (teacher)

- Ex 1a: 55° ; 12°
- Ex 1b: 135° ; 80°
- Ex 2a: $m = 62$

4. Ex 2b: $k = 133$
5. Ex 3a: $x = 29$
6. Ex 3b: 90° each

Easy (drill)

- | | | |
|----------------|-----------------|-----------------|
| 1. 76° | 12. 89° | 23. 21° |
| 2. 56° | 13. 73° | 24. 19° |
| 3. 53° | 14. 132° | 25. 15° |
| 4. 54° | 15. 103° | 26. 127° |
| 5. 24° | 16. 162° | 27. 141° |
| 6. 41° | 17. 179° | 28. 167° |
| 7. 84° | 18. 55° | 29. 99° |
| 8. 36° | 19. 106° | 30. 118° |
| 9. 67° | 20. 2° | 31. 67° |
| 10. 70° | 21. 126° | 32. 90° |
| 11. 10° | 22. 14° | |

Medium (drill)

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|-------|--------------|---------------|
| 1. C | 12. S | 23. $k = 23$ |
| 2. N | 13. N | 24. $m = 13$ |
| 3. C | 14. S | 25. $x = 50$ |
| 4. N | 15. S | 26. $m = 64$ |
| 5. C | 16. N | 27. $p = 46$ |
| 6. C | 17. C | 28. $k = 75$ |
| 7. S | 18. S | 29. $x = 87$ |
| 8. C | 19. N | 30. $x = 117$ |
| 9. S | 20. N | 31. $y = 53$ |
| 10. S | 21. $x = 94$ | 32. $k = 107$ |
| 11. C | 22. $x = 28$ | |

Challenging

1. 55° and 35°
2. 36°
3. $x = 40$

Extension

1. 60°
2. 40°

Exit ticket

1. Level 1: 28°
2. Level 2: 62°
3. Level 3: $x = 45$

2.05 Angles at a point and vertically opposite angles (booklet p. 27)

WE DO examples (teacher)

1. Ex 1a: $a = 110$
2. Ex 1b: $m = 25$
3. Ex 2a: $a = 130$
4. Ex 2b: $x = 120$
5. Ex 3a: $p = 145$
6. Ex 3b: $p = 155$, $q = 25$, $r = 155$

Easy (drill)

- | | | |
|---------------|---------------|---------------|
| 1. $a = 70$ | 11. $a = 84$ | 21. $a = 188$ |
| 2. $a = 99$ | 12. $a = 128$ | 22. $a = 202$ |
| 3. $a = 150$ | 13. $a = 103$ | 23. $a = 62$ |
| 4. $a = 102$ | 14. $a = 132$ | 24. $a = 221$ |
| 5. $a = 159$ | 15. $a = 202$ | 25. $a = 123$ |
| 6. $a = 164$ | 16. $a = 85$ | 26. $a = 155$ |
| 7. $a = 162$ | 17. $a = 217$ | 27. $a = 179$ |
| 8. $a = 154$ | 18. $a = 28$ | 28. $a = 169$ |
| 9. $a = 39$ | 19. $a = 174$ | |
| 10. $a = 116$ | 20. $a = 26$ | |

Medium (drill)

1. $v = 77^\circ$, $n = 103^\circ$
2. $v = 84^\circ$, $n = 96^\circ$

- | | | |
|-------------------------------------|-------------------------------------|--------------------------------------|
| 3. $v = 76^\circ$, $n = 104^\circ$ | 7. $v = 34^\circ$, $n = 146^\circ$ | 10. $v = 61^\circ$, $n = 119^\circ$ |
| 4. $v = 127^\circ$, $n = 53^\circ$ | 8. $v = 74^\circ$, $n = 106^\circ$ | 11. $v = 91^\circ$, $n = 89^\circ$ |
| 5. $v = 130^\circ$, $n = 50^\circ$ | 9. $v = 70^\circ$, $n = 110^\circ$ | 12. $v = 136^\circ$, $n = 44^\circ$ |
| 6. $v = 83^\circ$, $n = 97^\circ$ | | |

Challenging

1. $x = 132$ (straight line), $y = 48$ (vertically opposite)
2. $t = 173$ (angles at a point)

Extension

1. 30° , 60° , 90°
2. $x = 30$

Exit ticket

1. Level 1: 116
2. Level 2: 38°
3. Level 3: 135

2.06 Parallel and perpendicular lines

(booklet p. 33)

WE DO examples (teacher)

1. Ex 1a: $AB \parallel DC$ (or $AD \parallel BC$); $AB \perp BC$
2. Ex 1b: AB is perpendicular to CD ; EF is parallel to GH
3. Ex 2a: construction
4. Ex 2b: construction
5. Ex 3a: construction
6. Ex 3b: e.g. shelves ($\parallel$), wall and floor ($\perp$)

Easy (drill)

- | | | |
|----------------------|-------------|-------|
| 1. $PQ \perp RS$ | 7. $\perp$ | 13. P |
| 2. $\perp$ | 8. $\perp$ | 14. P |
| 3. $TU \perp VW$ | 9. P | 15. P |
| 4. $\perp$ | 10. P | 16. N |
| 5. $XY \parallel ZW$ | 11. $\perp$ | 17. N |
| 6. $KL \parallel MN$ | 12. $\perp$ | 18. N |

Medium (drill)

- | | | |
|----------|----------|-----------|
| 1. true | 5. true | 9. false |
| 2. true | 6. true | 10. true |
| 3. true | 7. true | 11. true |
| 4. false | 8. false | 12. false |

Challenging

1. construction
2. square construction

Extension

1. construction
2. every point on it is the same distance from A and B

Exit ticket

1. Level 1: $\parallel$
2. Level 2: e.g. wall and floor
3. Level 3: construction

2.07 Corresponding angles (booklet p. 39)

WE DO examples (teacher)

1. Ex 1a: bottom line, upper right
2. Ex 1b: $a = 65$
3. Ex 2a: $m = 110$
4. Ex 2b: the angles sit in an F
5. Ex 3a: $x = 125$, $y = 55$
6. Ex 3b: Only when the lines are parallel

Easy (drill)

- | | | |
|---------------|----------------|---------------|
| 1. 89° | 2. 104° | 3. 23° |
|---------------|----------------|---------------|

- | | | |
|-----------------|-----------------|---------------|
| 4. 136° | 11. 64° | 18. $a = 80$ |
| 5. 93° | 12. 121° | 19. $a = 65$ |
| 6. 55° | 13. $a = 90$ | 20. $a = 150$ |
| 7. 100° | 14. $a = 145$ | 21. $a = 65$ |
| 8. 44° | 15. $a = 65$ | 22. $a = 145$ |
| 9. 28° | 16. $a = 100$ | 23. $a = 45$ |
| 10. 143° | 17. $a = 140$ | 24. $a = 30$ |

Medium (drill)

- | | | |
|-----------------------|-----------------------|---------|
| 1. $a = 70, b = 110$ | 11. $a = 125, b = 55$ | 21. no |
| 2. $a = 85, b = 95$ | 12. $a = 40, b = 140$ | 22. no |
| 3. $a = 90, b = 90$ | 13. $x = 32$ | 23. yes |
| 4. $a = 150, b = 30$ | 14. $x = 14$ | 24. yes |
| 5. $a = 35, b = 145$ | 15. $x = 32$ | 25. no |
| 6. $a = 135, b = 45$ | 16. $x = 16$ | 26. no |
| 7. $a = 145, b = 35$ | 17. $x = 9$ | 27. yes |
| 8. $a = 60, b = 120$ | 18. $x = 27$ | 28. no |
| 9. $a = 105, b = 75$ | 19. $x = 11$ | |
| 10. $a = 30, b = 150$ | 20. $x = 16$ | |

Challenging

- a line crossing two or more lines
- drawing
- No

Extension

- $y = 40$
- $a = 30$

Exit ticket

- Level 1: 60
- Level 2: an F
- Level 3: $x = 20$

2.08 Alternate angles (booklet p. 45)

WE DO examples (teacher)

- Ex 1a: bottom line, upper right
- Ex 1b: $x = 60$
- Ex 2a: $k = 115$
- Ex 2b: True
- Ex 3a: $a = 48, b = 132$
- Ex 3b: the angles sit inside a Z

Easy (drill)

- | | | |
|----------------|-----------------|---------------|
| 1. 111° | 9. 113° | 17. $a = 140$ |
| 2. 28° | 10. 42° | 18. $a = 45$ |
| 3. 78° | 11. 133° | 19. $a = 40$ |
| 4. 119° | 12. 68° | 20. $a = 150$ |
| 5. 91° | 13. $a = 90$ | 21. $a = 115$ |
| 6. 75° | 14. $a = 135$ | 22. $a = 155$ |
| 7. 105° | 15. $a = 90$ | 23. $a = 75$ |
| 8. 139° | 16. $a = 85$ | 24. $a = 40$ |

Medium (drill)

- | | | |
|-----------------------------------|-----------------------------------|--------------|
| 1. $a = 83^\circ, c = 97^\circ$ | 11. $a = 30^\circ, c = 150^\circ$ | 21. $x = 27$ |
| 2. $a = 13^\circ, c = 167^\circ$ | | 22. $x = 12$ |
| 3. $a = 117^\circ, c = 63^\circ$ | 12. $a = 23^\circ, c = 157^\circ$ | 23. $x = 38$ |
| 4. $a = 43^\circ, c = 137^\circ$ | 13. $a = 51^\circ, c = 129^\circ$ | 24. $x = 19$ |
| 5. $a = 109^\circ, c = 71^\circ$ | 14. $a = 50^\circ, c = 130^\circ$ | 25. $x = 30$ |
| 6. $a = 52^\circ, c = 128^\circ$ | | 26. $x = 24$ |
| 7. $a = 62^\circ, c = 118^\circ$ | 15. $a = 39^\circ, c = 141^\circ$ | 27. $x = 27$ |
| 8. $a = 151^\circ, c = 29^\circ$ | 16. $x = 33$ | 28. $x = 39$ |
| 9. $a = 70^\circ, c = 110^\circ$ | 17. $x = 14$ | 29. $x = 9$ |
| 10. $a = 37^\circ, c = 143^\circ$ | 18. $x = 28$ | 30. $x = 18$ |
| | 19. $x = 9$ | 31. $x = 37$ |
| | 20. $x = 27$ | |

Challenging

- drawing
- $x = 45$
- alternate angles and angles on a straight line

Extension

- $a = 25$
- $m = 35$; both 105°

Exit ticket

- Level 1: 65
- Level 2: Z
- Level 3: $x = 35$

2.09 Co-interior angles (booklet p. 51)

WE DO examples (teacher)

- Ex 1a: bottom line, upper right
- Ex 1b: $y = 60$
- Ex 2a: $m = 105$
- Ex 2b: No: they add to 180° (unless both are 90°)
- Ex 3a: $p = 65, q = 115$
- Ex 3b: corresponding (equal), alternate (equal), co-interior (add to 180°)

Easy (drill)

- | | | |
|----------------|-----------------|---------------|
| 1. 31° | 9. 91° | 17. $a = 60$ |
| 2. 47° | 10. 157° | 18. $a = 100$ |
| 3. 79° | 11. 100° | 19. $a = 95$ |
| 4. 108° | 12. 137° | 20. $a = 90$ |
| 5. 74° | 13. $a = 145$ | 21. $a = 85$ |
| 6. 104° | 14. $a = 110$ | 22. $a = 65$ |
| 7. 112° | 15. $a = 30$ | 23. $a = 70$ |
| 8. 101° | 16. $a = 125$ | 24. $a = 95$ |

Medium (drill)

- | | | |
|--------------|--------------|---------|
| 1. $y = 48$ | 12. $y = 43$ | 23. no |
| 2. $y = 16$ | 13. $y = 8$ | 24. yes |
| 3. $y = 32$ | 14. $y = 51$ | 25. yes |
| 4. $y = 21$ | 15. $y = 5$ | 26. no |
| 5. $y = 50$ | 16. $y = 23$ | 27. no |
| 6. $y = 58$ | 17. yes | 28. no |
| 7. $y = 15$ | 18. no | 29. no |
| 8. $y = 55$ | 19. yes | 30. no |
| 9. $y = 49$ | 20. no | 31. yes |
| 10. $y = 54$ | 21. no | 32. yes |
| 11. $y = 40$ | 22. no | |

Challenging

- $x = 60$
- 90° each; it is perpendicular to the parallel lines
- drawing

Extension

- 80° and 100°
- $a = 50$

Exit ticket

- Level 1: 70
- Level 2: 180°
- Level 3: $y = 20$

2.10 Finding unknown angles on parallel lines (booklet p. 57)

WE DO examples (teacher)

- Ex 1a: $a = 65$ (corresponding then straight line; or co-interior with 115° pair)
- Ex 1b: $m = 35$ (alternate)
- Ex 2a: $x = 37$ (corr.), $y = 143$ (straight line)

4. Ex 2b: $k = 65$ (alternate)
5. Ex 3a: $p = 62$, $m = 62$, $z = 118$
6. Ex 3b: use two rules in steps, e.g. vertically opposite then alternate

Easy (drill)

- | | | |
|------------------|--------------|---------------|
| 1. co-interior | 6. alternate | 11. $a = 30$ |
| 2. co-interior | 7. $a = 100$ | 12. $a = 85$ |
| 3. alternate | 8. $a = 105$ | 13. $a = 85$ |
| 4. corresponding | 9. $a = 150$ | 14. $a = 150$ |
| 5. co-interior | 10. $a = 80$ | 15. $a = 135$ |

Medium (drill)

- | | | |
|-------------------------------------|--------------------------------------|-----------------------------|
| 1. $129^\circ, 51^\circ, 51^\circ$ | 7. $142^\circ, 38^\circ, 38^\circ$ | 13. $a = 155$ |
| 2. $52^\circ, 128^\circ, 128^\circ$ | 8. $18^\circ, 162^\circ, 162^\circ$ | (corresponding) |
| 3. $55^\circ, 125^\circ, 125^\circ$ | 9. $130^\circ, 50^\circ, 50^\circ$ | 14. $a = 40$ (co-interior) |
| 4. $31^\circ, 149^\circ, 149^\circ$ | 10. $16^\circ, 164^\circ, 164^\circ$ | 15. $a = 135$ (co-interior) |
| 5. $79^\circ, 101^\circ, 101^\circ$ | 11. $127^\circ, 53^\circ, 53^\circ$ | |
| 6. $104^\circ, 76^\circ, 76^\circ$ | 12. $141^\circ, 39^\circ, 39^\circ$ | |

Challenging

1. four of 57° and four of 123°
2. 135° and 45°
3. The lines are parallel so each angle has a matching corresponding angle

Extension

1. $x = 16$
2. $x = 40$

Exit ticket

1. Level 1: 58
2. Level 2: alternate (vertically opposite then corresponding)
3. Level 3: four 58° and four 122°

2.11 Proving lines are parallel (booklet p. 63)

WE DO examples (teacher)

1. Ex 1a: Yes: co-interior angles $135 + 45 = 180$
2. Ex 1b: No: $110 + 72 = 182$
3. Ex 2a: Yes: corresponding angles equal
4. Ex 2b: $x = 95$
5. Ex 3a: corresponding equal, alternate equal, co-interior supplementary

6. Ex 3b: we use the angle to prove the lines are parallel (the reverse of the rule)

Easy (drill)

- | | | |
|--------|---------|---------|
| 1. yes | 9. no | 17. yes |
| 2. yes | 10. no | 18. no |
| 3. yes | 11. yes | 19. yes |
| 4. yes | 12. no | 20. yes |
| 5. no | 13. yes | 21. no |
| 6. yes | 14. no | 22. no |
| 7. yes | 15. no | 23. yes |
| 8. no | 16. yes | 24. yes |

Medium (drill)

- | | | |
|---------------|--|--|
| 1. $x = 150$ | 14. $x = 125$ | 20. co-interior angles are supplementary |
| 2. $x = 45$ | 15. $x = 60$ | 21. alternate angles are equal |
| 3. $x = 80$ | 16. alternate angles are equal | 22. co-interior angles are supplementary |
| 4. $x = 125$ | 17. co-interior angles are supplementary | 23. alternate angles are equal |
| 5. $x = 85$ | 18. co-interior angles are supplementary | |
| 6. $x = 125$ | 19. co-interior angles are supplementary | |
| 7. $x = 35$ | | |
| 8. $x = 30$ | | |
| 9. $x = 120$ | | |
| 10. $x = 65$ | | |
| 11. $x = 35$ | | |
| 12. $x = 150$ | | |
| 13. $x = 40$ | | |

Challenging

1. alternate angles equal, so parallel
2. $108 + 72 = 180$, co-interior supplementary, so parallel
3. No: corresponding angles are not equal

Extension

1. $x = 132$; $132 + 48 = 180$, so yes (co-interior)
2. Measurements are never exact: a proof uses the angle rules

Exit ticket

1. Level 1: Yes
2. Level 2: No: $105 + 85 = 190$, not 180
3. Level 3: $x = 37$